Analogue Electronic Circuits 1

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Analogue Electronic Circuits 1 (A04588)

Short Title:	Analogue Electronic Circuits 1		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electronics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module follows on from Analogue Electronic Devices and builds on concepts introduced there. Equivalent circuit representations are used to help understanding of device/circuit operation under d.c. and a.c. conditions. SCR based power control circuits are developed. Operational amplifiers are introduced for linear circuit applications and sine wave oscillation theory is developed.

Programmes

`ELTR-0013`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	2 / 3 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	2 / 3 / M

Indicative Content

- Diode Circuit Design and Application
- BJT Circuit Design and Application
- BJT Circuit Models
- FET Circuit Design and Application
- FET Circuit Models
- MOSFET Switching Circuits
- Op-Amp Circuit Design and Application
- SCR Circuit Design and Application
- Sine Wave Oscillator Theory & Wien Bridge Circuits

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Analyse typical diode circuits.
2. Demonstrate a core knowledge of BJT and FET based amplifiers and their d.c./a.c equivalent circuits.
3. Differentiate between the ideal and non-ideal characteristics of operational amplifiers.
4. Develop fundamental linear operational amplifier circuit designs.
5. Analyse typical SCR based power control circuits.
6. Explain fundamental oscillator theory.
7. Perform practical tasks such as building, testing and documenting analogue circuits.

Learning and Teaching Methods

- Lectures
- Integrated Tutorials
- Laboratory Work
- Project Work

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5,6
Continuous Assessment	30%	
Lab Report	20%	7
Project	10%	7

Assessment Criteria

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Bogart, T., J.S. Beasley and G. Rico. _Electronic Devices and Circuits_. USA: Pearson, 2004.
- Malvino, A.P. and D. Bates. _Electronic Principles_. 8th ed. USA: McGraw Hill, 2015.

Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Electronics